Matthew J. Liu

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Professional Summary

Chemical engineer with deep expertise in the physical sciences, applying my scientific training and first-principles reasoning toward machine learning research.

Professional Experience

Stealth Startup — San Francisco, CA
ML Researcher — Nov 2026 – Present

Lawrence Berkeley National Laboratory — Berkeley, CA
Arnold O. Beckman Postdoctoral Fellow in Chemical Sciences — Sep 2025 – Nov 2026
Advisor: Prof. Kristin Persson (Director, Materials Project)

- Adapted materials-informatics methods into a self-supervised transformer model for rock geochemistry, with calibrated probabilistic predictions for mineralogy
- Extended the Materials Project database to disordered materials through a cluster expansion-driven DFT workflow that resolved ground-state structures and energies from a targeted set of r2SCAN calculations

Engine Ventures — Cambridge, MA
Technology Venture Fellow — Apr 2025 – Sep 2025
Advisor: Dr. Richard Collins

- Conducted technical due diligence by developing comprehensive views of scientific research and technical risk within the fields of AI for science and environmental remediation

Massachusetts Institute of Technology — Cambridge, MA
Arnold O. Beckman Postdoctoral Fellow in Chemical Sciences — Sep 2024 – Aug 2025
Advisor: Prof. Yuriy Roman (Dept. Chemical Engineering)

- Investigated electric field-promoted heterogeneous Brønsted acid catalysis by iterating on catalyst synthesis, chemical analysis, and custom reactor development

X, The Moonshot Factory — Mountain View, CA
PhD Resident — Oct 2023 – Feb 2024
Advisors: Dr. Shijian Jin and Uri Kelly

- Developed novel electrochemical processes by rapidly prototyping custom experimental rigs and performing electroanalytical measurements
- Built electrochemical simulations of coupled transport and reaction to guide process development

Stanford University — Stanford, CA
NASA NSTGRO | ARCS Research Fellow — Apr 2019 – Aug 2024
Advisor: Prof. William Tarpeh (Dept. Chemical Engineering)

- Electrochemical separations and electrocatalysis for resource recovery

Education

Stanford University, Stanford, CA — 2018 – 2024
Ph.D., Chemical Engineering
Advisor: Prof. William Tarpeh (Dept. Chemical Engineering)

University of California, Berkeley, Berkeley, CA — 2014 – 2018
B.S. with Honors, Chemical Engineering

Selected Publications

M.J. Liu, D.M. Miller, U.V. Iwata, W.A. Tarpeh. Tuning electrolyte composition to modulate catalytic performance in molecularly catalyzed electrochemical nitrate reduction. *ACS Electrochemistry* 2025, 1 (10), 2249–2263. [Link]

M.J. Liu, C.A. Fernández Otero, D. Uruchurtu Patino, H. Gong, M.D. Hossain, J.E. Matthews, K.S. Williams, A. Vargas, M.J. Zachman, A.S. Hoffman, D. Nordlund, M. Bajdich, S.R. Bare, M.B. Stevens, T.F. Jaramillo, Z. Bao, W.A. Tarpeh. Titanium-, nitrogen-doped carbon flowers catalyze electrochemical nitrate reduction reaction to ammonia. *Journal of the American Chemical Society* 2025, 147 (32), 29026–29041. [Link]

M.J. Liu*, J. Guo*, A.S. Hoffman, J.H. Stenlid, M.T. Tang, E.R. Corson, K.H. Stone, F. Abild-Pedersen, S.R. Bare, W.A. Tarpeh. Catalytic performance and near-surface X-ray characterization of titanium hydride electrodes for the electrochemical nitrate reduction reaction. *Journal of the American Chemical Society* 2022, 144 (13), 5739–5744. Featured in JACS Spotlights. [Link]

M.J. Liu, B.S. Neo, W.A. Tarpeh. Building an operational framework for selective nitrogen recovery via electrochemical stripping. *Water Research* 2020, 169, 115226. [Link]

*Equal contribution

Workshop Papers

M.J. Liu, A.D. Kaplan, M.L. Whittaker, K.A. Persson. Simulated petrological labels enable probabilistic mineralogy prediction from whole-rock geochemistry. *Sim2Science Workshop, NeurIPS* 2026. arXiv:2610.36429. [Link]

M.J. Liu, W.H. Zheng, V. Purohit, S. Xie, C. Li, N. Flynn. Scaling laws for grid-based approximate nearest neighbor search in high dimensions. *High-dimensional Learning Dynamics Workshop, ICML* 2026. arXiv:2607.01283. [Link]

Patents

D.M. Miller, W.A. Tarpeh, **M.J. Liu**, K. Abels, K. Chan, K.S. Williams, A. Kogler, J. Guo, B.D. Clark. Molecular electrocatalytic systems and methods thereof. US Patent App. PCT/US2024/042321 (2024).

W.A. Tarpeh, A. Kogler, B.D. Clark, **M.J. Liu**, W. Chow. Systems and methods of flexible electrochemical stripping to recover alkaline ammonia and acidic ammonium from wastewaters. US Patent App. 18/041,678 (2024).

W.A. Tarpeh, **M.J. Liu**, A. Kogler, K.S. Williams, J. Simon, R. Wood. Annular electrochemical cell for stripping and recovering ammonia from wastewaters. Stanford Docket Number S23-303 (2023).

W.A. Tarpeh, **M.J. Liu**, B.D. Clark. Apparatus and methods for nitrogen reduction in wastewater. US Patent App. PCT/US2021/046015 (2020).

Selected Honors and Awards

Engine Ventures Fellowship	2025
Arnold O. Beckman Postdoctoral Fellowship in Chemical Sciences	2024
ACS Graduate Student Award in Environmental Chemistry	2023
Achievement Rewards for College Scientists (ARCS) Fellowship	2022
ECS IE&EE Student Achievement Award	2022
NASA Space Technology Graduate Research Opportunities Fellowship	2020
Chevron Design Challenge, 1st place	2018
SCI Scholar	2017
Cal Energy Corps Scholar	2015

Skills

Machine Learning

Models: self-supervised pretraining, transformer architectures, probabilistic modeling, calibration and uncertainty quantification

Inference: sim-to-real transfer from simulation-generated training data, simulation-based inference with bootstrap uncertainty

Systems: LLM agents with tool use for scientific inference, approximate nearest neighbor search

Simulation & Scientific Computing

Atomistic: high-throughput density functional theory (VASP, r2SCAN, pymatgen), cluster expansions, machine-learned interatomic potentials (MACE, CHGNet)

Continuum and statistical: thermodynamic modeling (MAGEMin), finite element modeling (FEniCS), Monte Carlo simulation, optimal experimental design

Software & Infrastructure

Languages: Python (PyTorch, NumPy, pandas), C++ (Armadillo, HDF5/HighFive)

Compute: HPC (NERSC Perlmutter GPU nodes, Slurm), distributed computing (Dask)

Data and workflows: workflow orchestration (atomate2, jobflow, FireWorks), MongoDB, S3 object storage, testing and reproducible pipelines

Experimental R&D

Electrochemistry: voltammetry, chronoamperometry, galvanostatic cycling, controlled-potential electrolysis, rotating disk electrode, impedance spectroscopy, Tafel and foot-of-the-wave analysis

Catalysis: heterogeneous and molecular catalysis, reaction engineering, catalyst synthesis (sol-gel, calcination, wet impregnation), process development

Materials characterization: XPS, XRD, XAS, ToF-SIMS, SEM, TEM, EDS, ion chromatography, GC-FID, GC-MS, NMR, isotope labeling